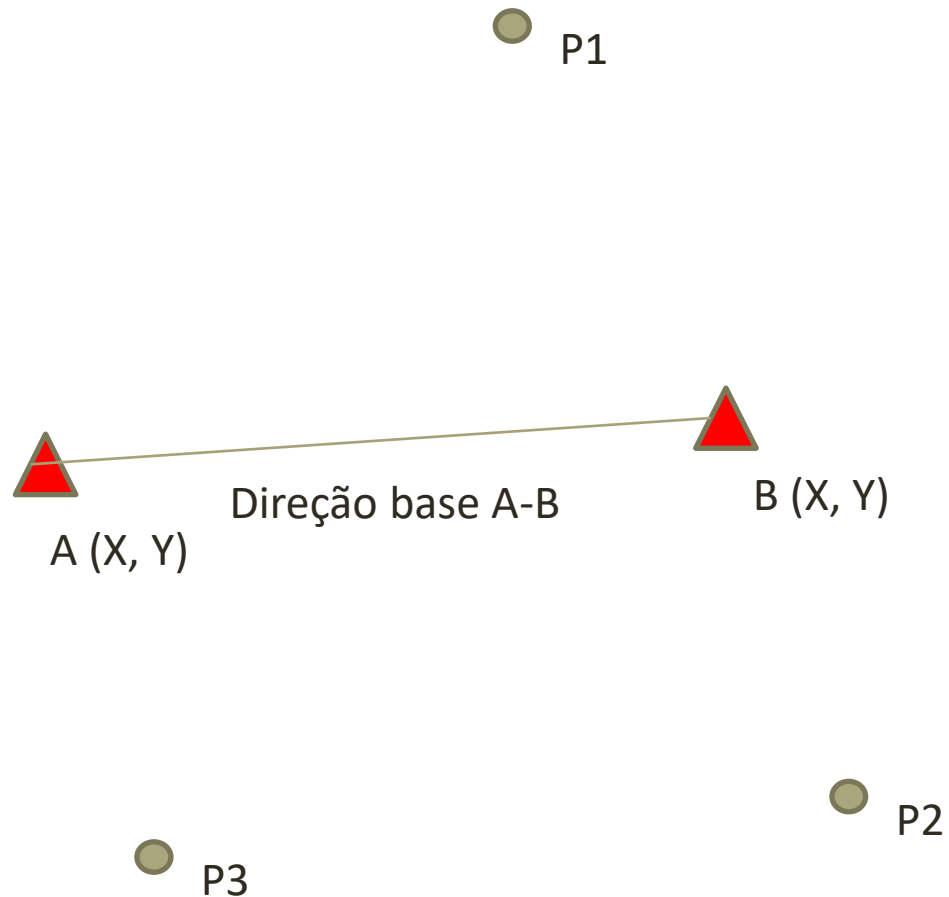


Métodos de Levantamento Planimétrico

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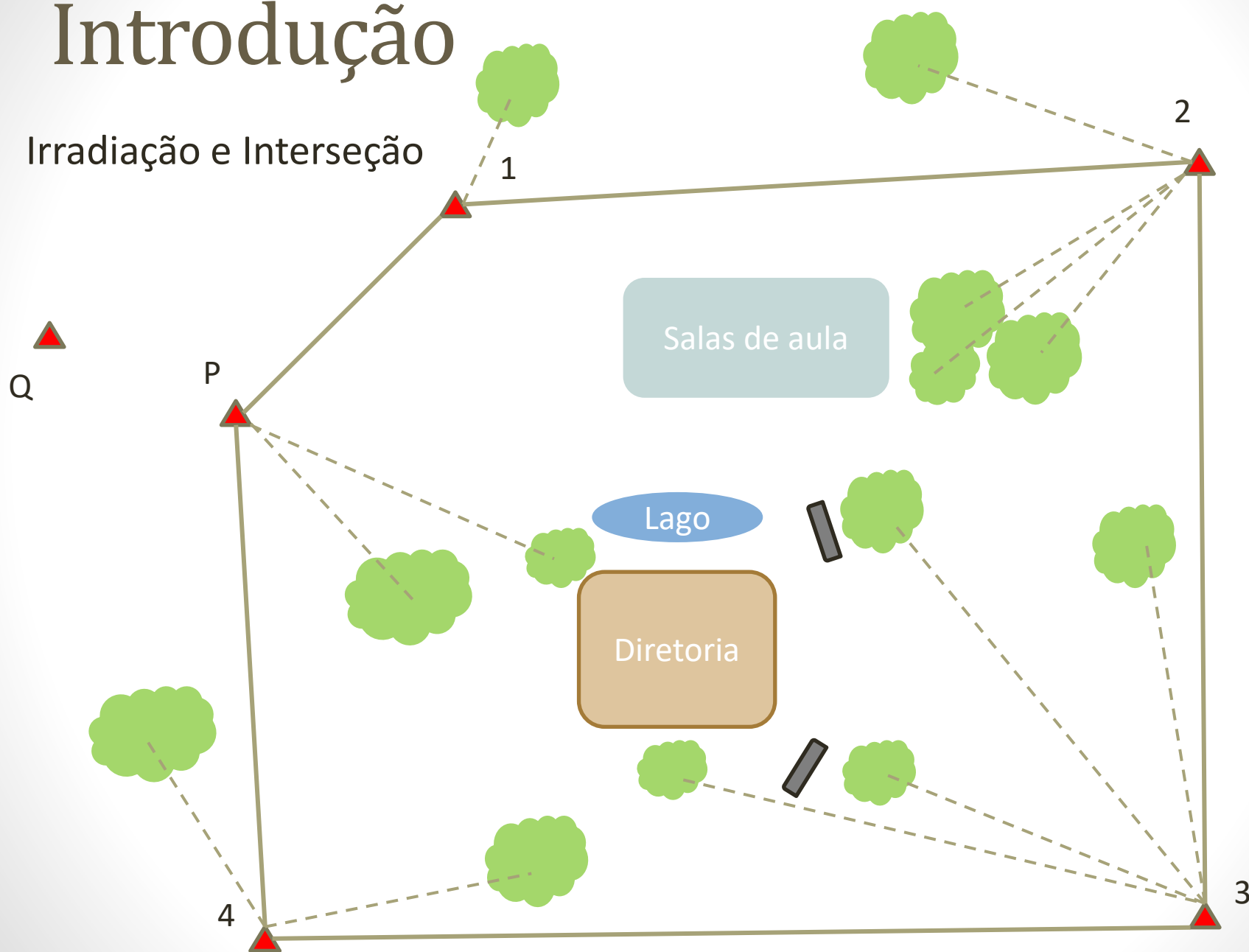
Introdução

- Irradiação e Interseção

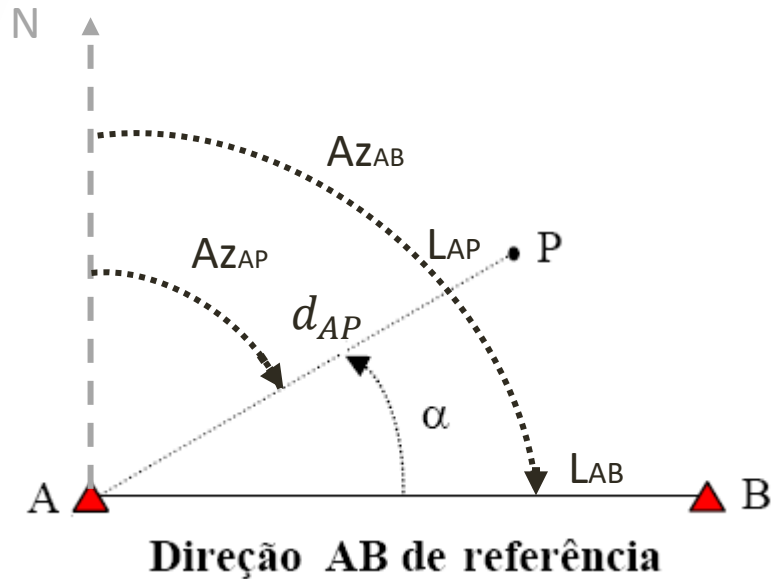


Introdução

Irradiação e Interseção



Irradiação



Dados:

A (X, Y)

B (X, Y)

Medidos:

L_{AB} e L_{AP}

d_{AP}

$$X_P = X_A + \Delta X_{AP}$$

$$Y_P = Y_A + \Delta Y_{AP}$$

$$\Delta X_{AP} = d_{AP} \cdot \text{sen} Az_{AP}$$

$$\Delta Y_{AP} = d_{AP} \cdot \text{cos} Az_{AP}$$

Irradiação

$$AZ_i = AZ_{i-1} + \alpha_i - 180^\circ$$

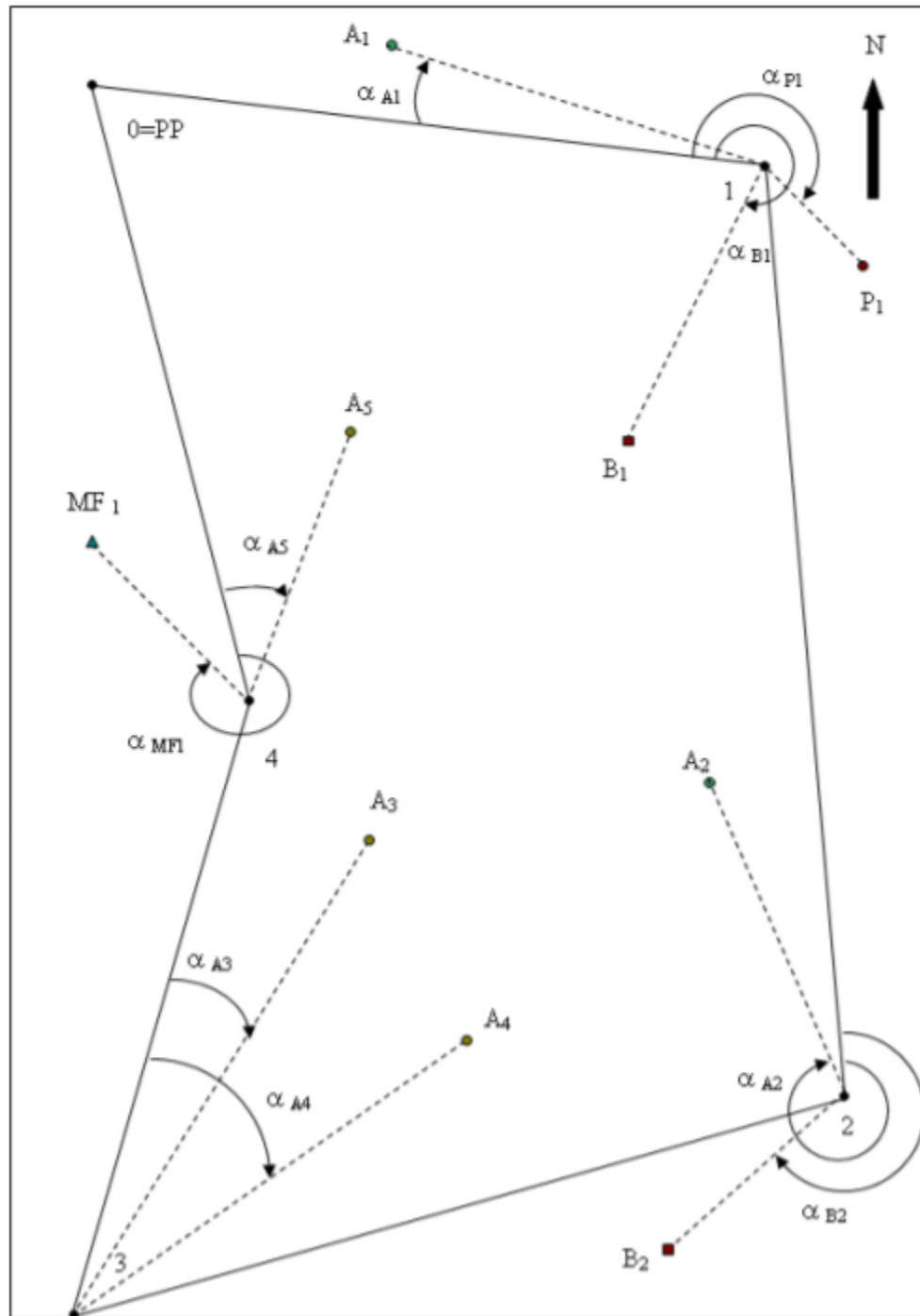
$$\Delta X = d \cdot \sin AZ$$

$$\Delta Y = d \cdot \cos AZ$$

$$X_i = X_{i-1} + \Delta X$$

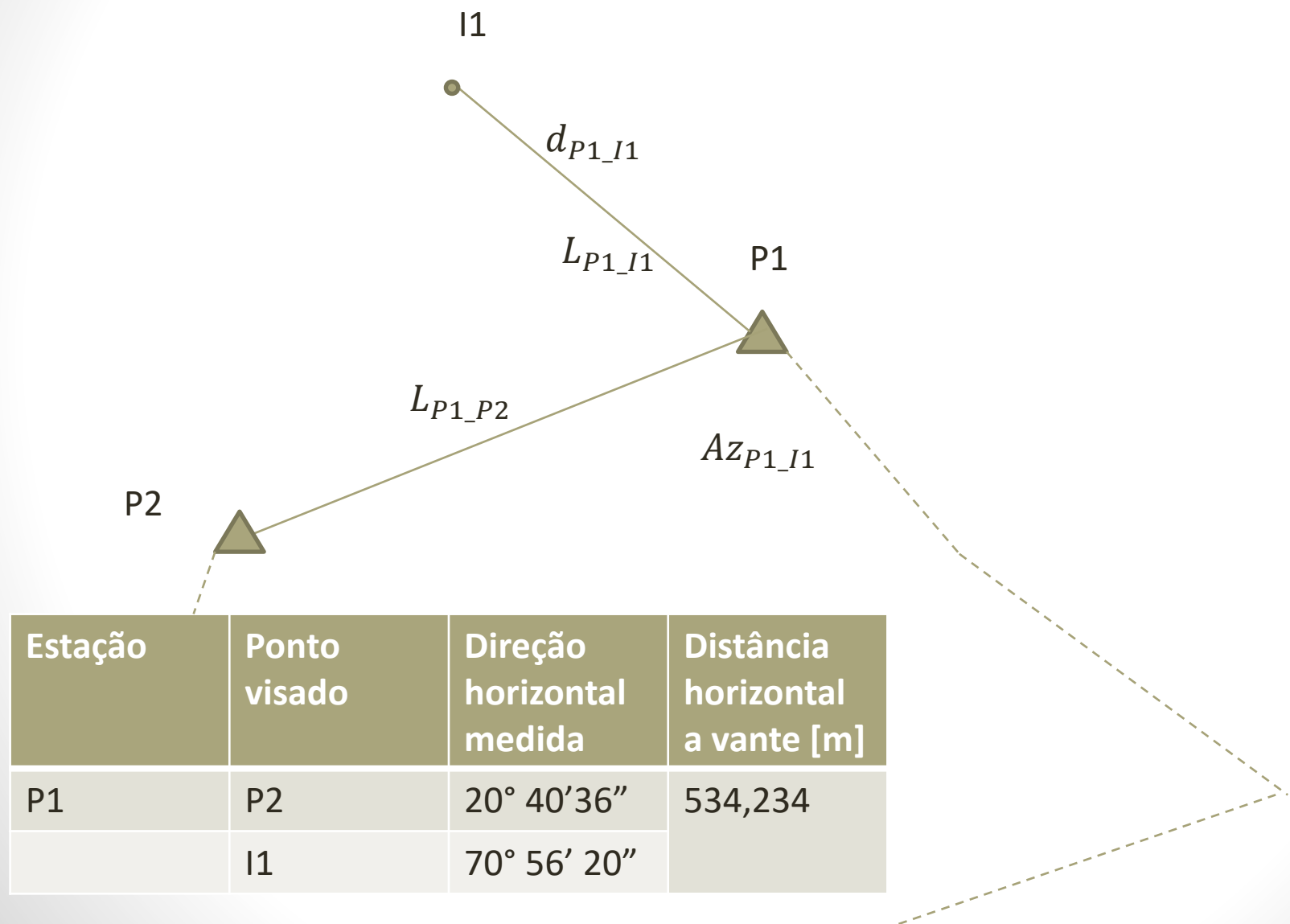
$$Y_i = Y_{i-1} + \Delta Y$$

Obs.: Primeiro as coordenadas dos vértices pertencentes a Poligonal Fechada devem ser calculadas



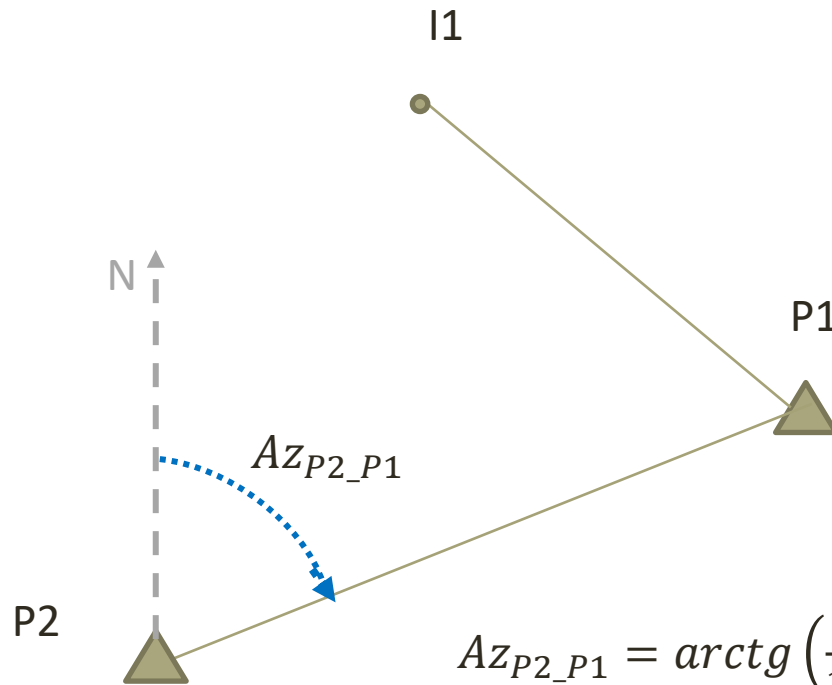
Exemplo

Ponto	X (m)	Y(m)
P1	7453,743	12743,125
P2	6676,216	11633,531



Exemplo

Ponto	X (m)	Y(m)
P1	7453,743	12743,125
P2	6676,216	11633,531



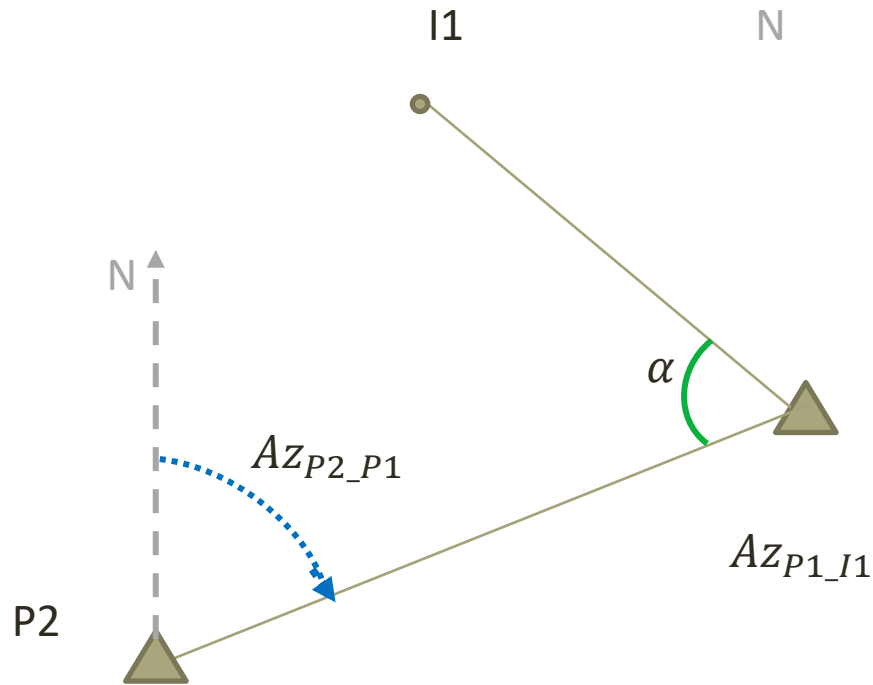
1. Cálculo do Azimute Az_{P2_P1}

$$Az_{P2_P1} = \arctg \left(\frac{7453,743 - 6676,216}{12743,125 - 11633,531} \right) = 35^\circ 1' 12,44''$$

Estação	Ponto visado	Direção horizontal medida	Distância horizontal a vante [m]
P1	P2	20° 40' 36"	534,234
	I1	70° 56' 20"	

Exemplo

Ponto	X (m)	Y(m)
P1	7453,743	12743,125
P2	6676,216	11633,531



$$Az_{P2_P1} = 35^{\circ} 1' 12,44''$$

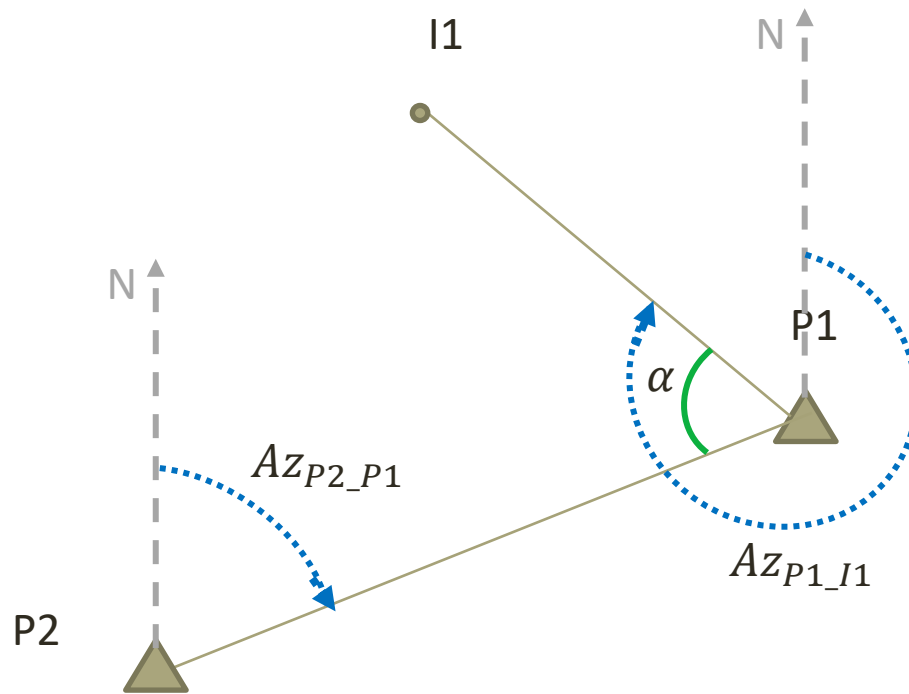
2. Cálculo do ângulo horizontal α

$$\alpha = 50^{\circ} 15' 44''$$

Estação	Ponto visado	Direção horizontal medida	Distância horizontal a vante [m]
P1	P2	20° 40' 36"	534,234
	I1	70° 56' 20"	

Exemplo

Ponto	X (m)	Y(m)
P1	7453,743	12743,125
P2	6676,216	11633,531



$$AZ_{P2_P1} = 35^{\circ} 1' 12,44''$$

$$\alpha = 50^{\circ} 15' 44''$$

3. Cálculo do Azimute AZ_{P1_I1}

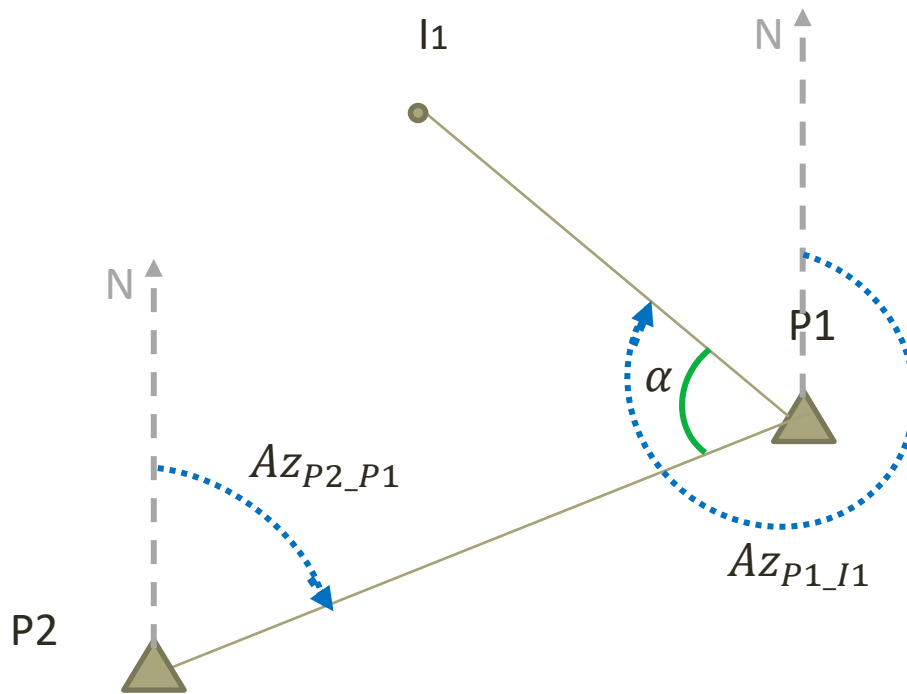
$$AZ_{P1_I1} = AZ_{P2_P1} + \alpha - 180^{\circ}$$

$$AZ_{P1_I1} = 265^{\circ} 16' 56,4''$$

Estação	Ponto visado	Direção horizontal medida	Distância horizontal a vante [m]
P1	P2	$20^{\circ} 40' 36''$	534,234
	I1	$70^{\circ} 56' 20''$	

Exemplo

Ponto	X (m)	Y(m)
P1	7453,743	12743,125
P2	6676,216	11633,531



$$AZ_{P1-I1} = 265^{\circ}16'59,4''$$

4. Cálculo da coordenadas do ponto I1

$$\Delta X = d. \sin Az$$

$$\Delta Y = d. \cos Az$$

$$\Delta X = -532,425 \text{ m}$$

$$\Delta Y = -43,931 \text{ m}$$

$$X_{I1} = X_{P1} + \Delta X$$

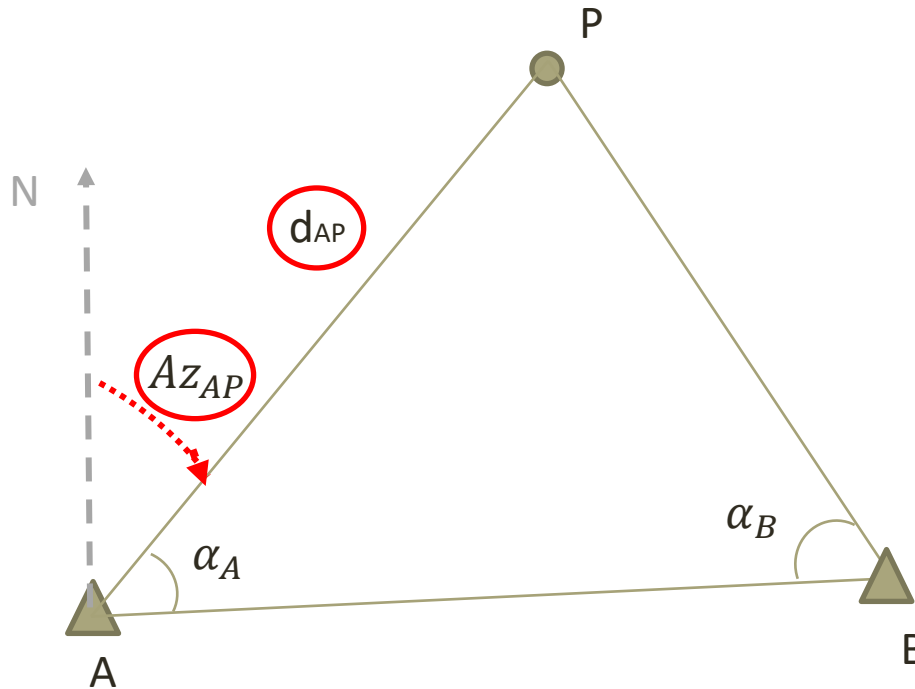
$$= 6921,318 \text{ m}$$

$$Y_{I1} = Y_{P1} + \Delta Y$$

$$= 12699,194 \text{ m}$$

Estação	Ponto visado	Direção horizontal medida	Distância horizontal a vante [m]
P1	P2	20° 40' 36"	534,234
	I1	70° 56' 20"	

Interseção



Dados:

Ponto A : (X, Y)

Ponto B: (X, Y)

Medidos:

L_{AP} , L_{AB} , L_{BA} , L_{BP}

Calcular:

Coordenadas de P (X, Y)

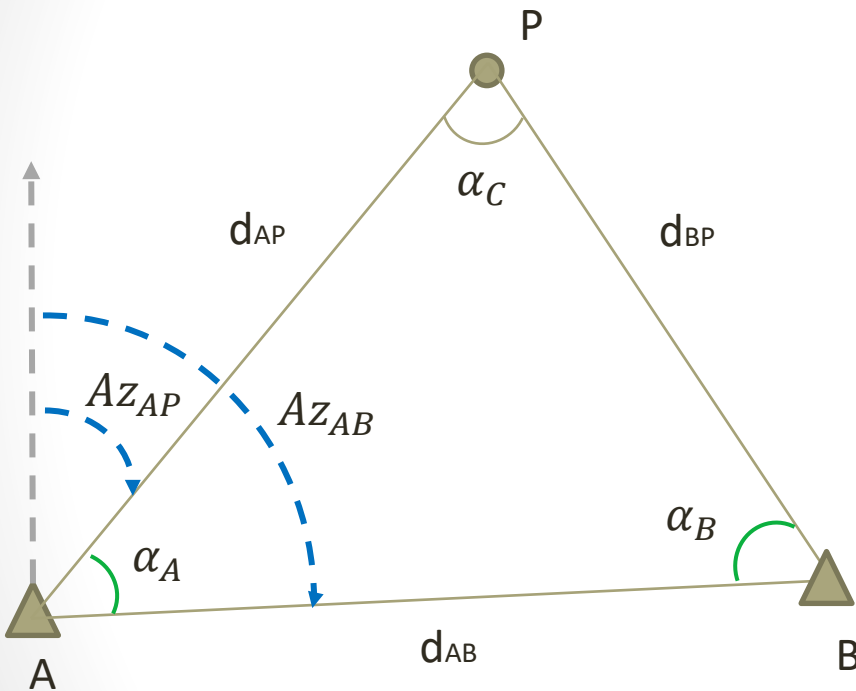
$$X_P = X_A + \Delta X_{AP}$$

$$Y_P = Y_A + \Delta Y_{AP}$$

$$\Delta X_{AP} = d_{AP} \cdot \text{sen} Az_{AP}$$

$$\Delta Y_{AP} = d_{AP} \cdot \text{cos} Az_{AP}$$

Interseção



1. Calcular Azimute da direção A-B (Az_{AB})
2. Calcular ângulos internos α_A e α_B
3. Calcular azimuth $Az_{AP} = Az_{AB} - \alpha_A$
4. Calcular a distância d_{AB}

$$d_{AB} = \sqrt{(X_B - X_A)^2 + (Y_B - Y_A)^2}$$

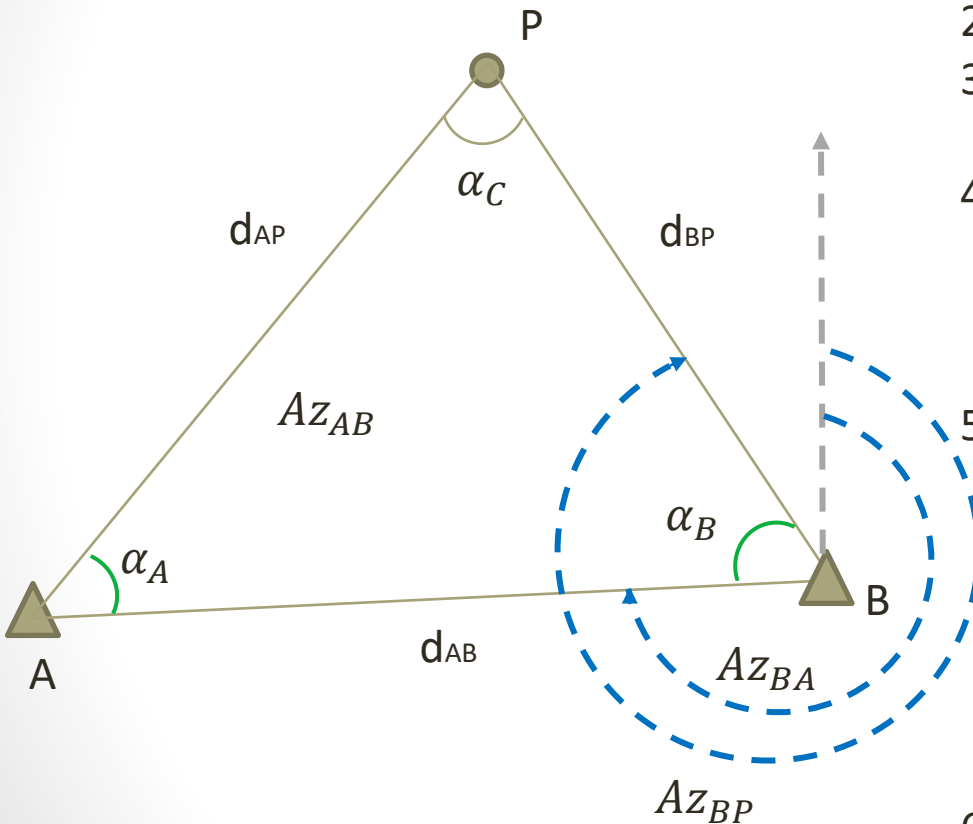
5. Calcular distância d_{AP}

$$\frac{d_{AP}}{\text{sen} \alpha_B} = \frac{d_{AB}}{\text{sen} \alpha_C}$$

$$\alpha_A + \alpha_B + \alpha_C = 180^\circ$$

6. Cálculo das coordenadas de P

Interseção



1. Calcular Azimute da direção B-A (Az_{BA})
2. Calcular ângulos internos α_A e α_B
3. Calcular azimuth $Az_{BP} = Az_{BA} + \alpha_B$
4. Calcular a distância d_{AB}

$$d_{AB} = \sqrt{(X_B - X_A)^2 + (Y_B - Y_A)^2}$$

5. Calcular distância d_{BP}

$$\frac{d_{BP}}{\text{sen} \alpha_A} = \frac{d_{AB}}{\text{sen} \alpha_C}$$

$$\alpha_A + \alpha_B + \alpha_C = 180^\circ$$

6. Cálculo das coordenadas de P

Exemplo

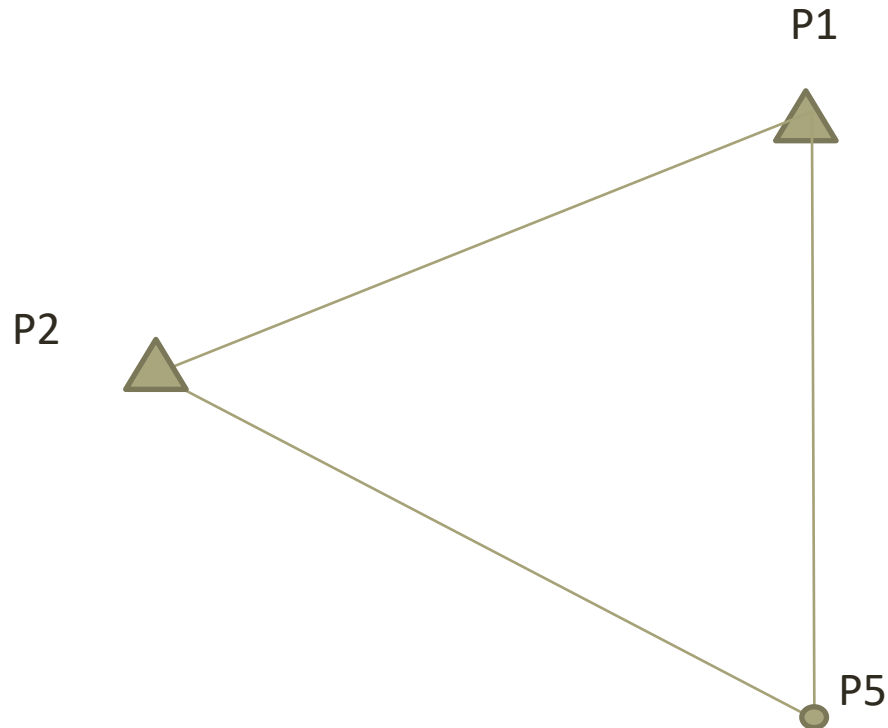
Determinar as coordenadas do P5

Estação	Ponto visado	Direção horizontal medida
P1	P2	0° 0' 0"
	P5	323° 56' 0"
P2	P1	0° 0' 0"
	P5	38° 59' 36"

Ponto	X (m)	Y(m)
P1	7453,743	12743,125
P2	6676,216	11633,531

Exemplo

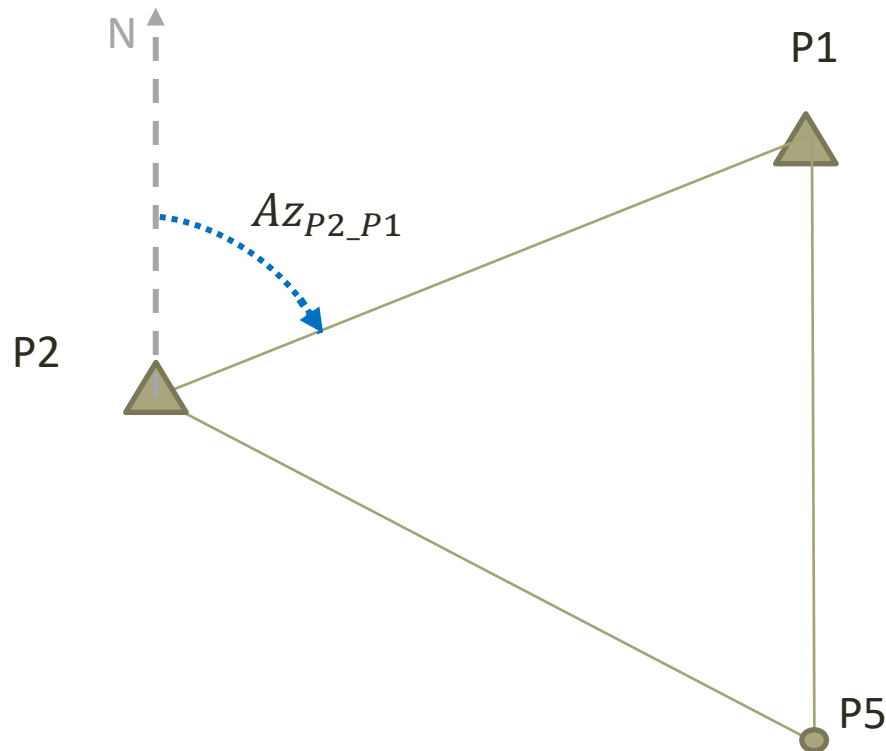
Ponto	X (m)	Y(m)
P1	7453,743	12743,125
P2	6676,216	11633,531



Estação	Ponto visado	Direção horizontal medida
P1	P2	0° 0' 0"
	P5	323° 56' 0"
P2	P1	0° 0' 0"
	P5	38° 59' 36"

Exemplo

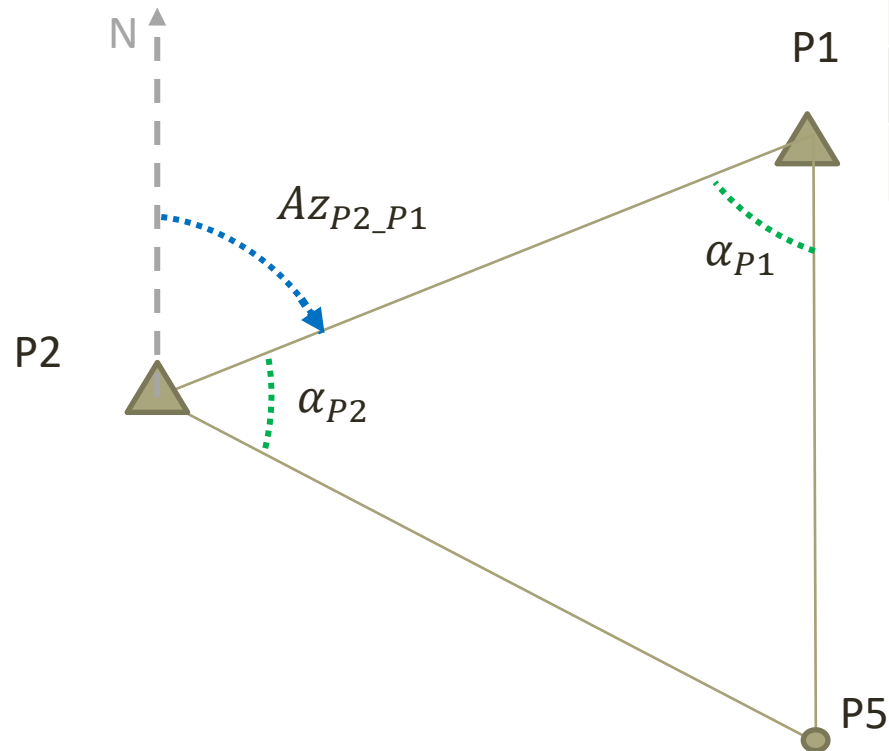
Ponto	X (m)	Y(m)
P1	7453,743	12743,125
P2	6676,216	11633,531



1. Calcular Azimute da direção P2-P1 (Az_{P2_P1})

$$Az_{P2_P1} = \arctg \left(\frac{7453,743 - 6676,216}{12743,125 - 11633,531} \right) = 35^\circ 1' 12,44''$$

Exemplo



Estação	Ponto visado	Direção horizontal medida
P1	P2	$0^{\circ} 0' 0''$
	P5	$323^{\circ} 56' 0''$
P2	P1	$0^{\circ} 0' 0''$
	P5	$38^{\circ} 59' 36''$

2. Calcular ângulos internos α_{P1} e α_{P2}

$$\alpha_{P1} = 0^{\circ} 0' 0'' - 323^{\circ} 56' 0'' = -323^{\circ} 56' 0'' + 360^{\circ} = 36^{\circ} 4' 0''$$

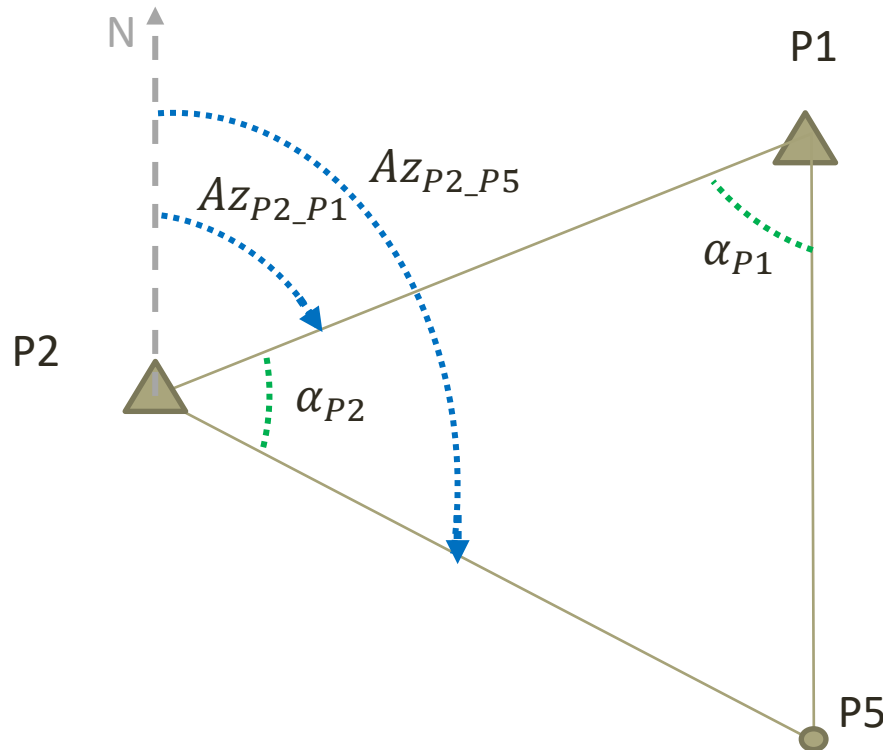
$$\alpha_{P2} = 38^{\circ} 59' 36'' - 0^{\circ} 0' 0'' = 38^{\circ} 59' 0''$$

Exemplo

$$Az_{P2_P1} = 35^{\circ} 1' 12,44''$$

$$\alpha_{P1} = 36^{\circ} 4' 0''$$

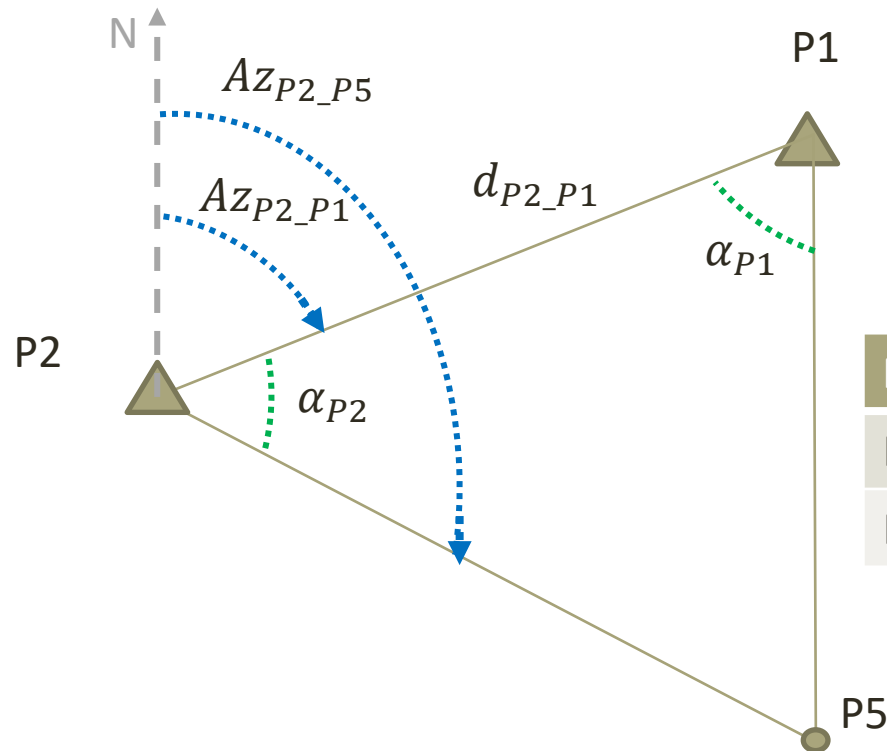
$$\alpha_{P2} = 38^{\circ} 59' 36''$$



3. Calcular azimuth $Az_{P2_P5} = Az_{P2_P1} + \alpha_{P2}$

$$Az_{P2_P5} = Az_{P2_P1} + \alpha_{P2} = 35^{\circ} 1' 12,44'' + 38^{\circ} 59' 36'' = 74^{\circ} 0' 48,44''$$

Exemplo



$$Az_{P2_P1} = 35^{\circ} 1' 12,44''$$

$$\alpha_{P1} = 36^{\circ} 4' 0''$$

$$\alpha_{P2} = 38^{\circ} 59' 0''$$

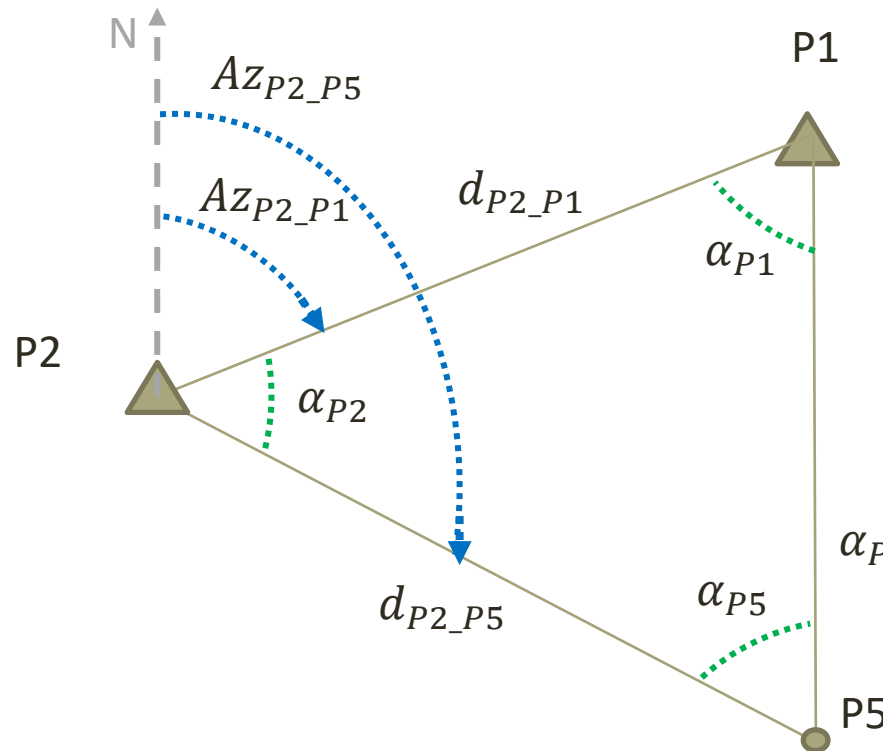
$$Az_{P2_P5} = 74^{\circ} 0' 48,44''$$

Ponto	X (m)	Y(m)
P1	7453,743	12743,125
P2	6676,216	11633,531

4. Calcular a distância d_{P2_P1}

$$d_{P2_P1} = \sqrt{(X_{P1} - X_{P2})^2 + (Y_{P1} - Y_{P2})^2} = 1354,897 \text{ m}$$

Exemplo



$$Az_{P2_P1} = 35^{\circ} 1' 12,44''$$

$$\alpha_{P1} = 36^{\circ} 4' 0''$$

$$\alpha_{P2} = 38^{\circ} 59' 36''$$

$$Az_{P2_P5} = 74^{\circ} 0' 12,44''$$

$$d_{P2_P1} = 1354,897 \text{ m}$$

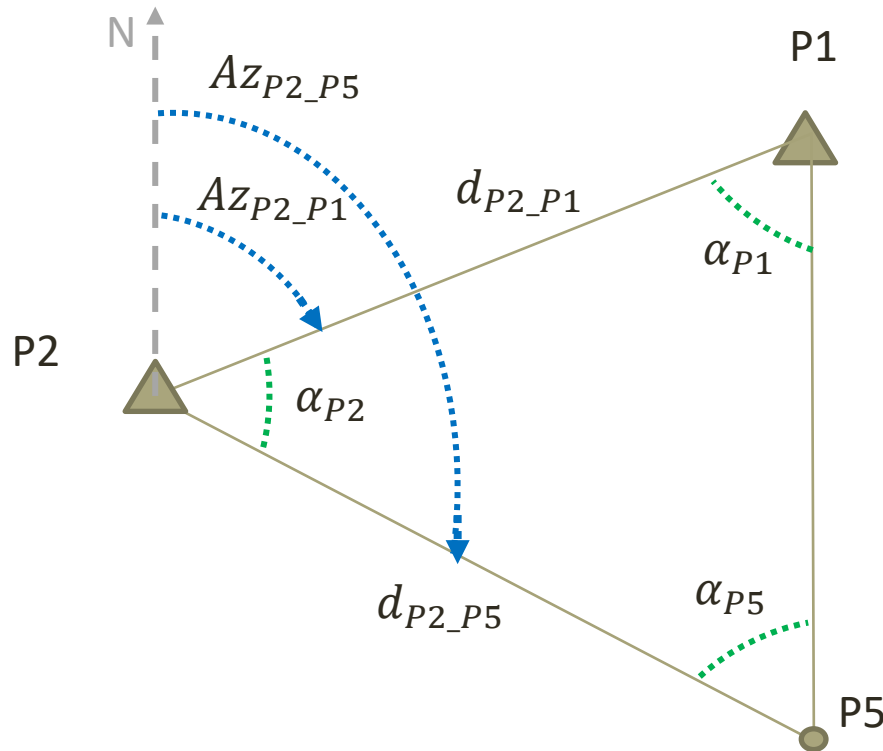
$$\alpha_{P5} = 180^{\circ} - 36^{\circ} 4' - 38^{\circ} 59' 36''$$

$$\alpha_{P5} = 104^{\circ} 56' 24''$$

5. Calcular distância d_{P2_P5}

$$\frac{d_{P2_P5}}{\text{sen } \alpha_{P1}} = \frac{d_{P2_P1}}{\text{sen } \alpha_{P5}} \quad d_{P2_P5} = \frac{d_{P2_P1} * \text{sen } \alpha_{P1}}{\text{sen } \alpha_{P5}} = 825,571 \text{ m}$$

Exemplo



$$Az_{P2_P1} = 35^\circ 1' 12,44''$$

$$\alpha_{P1} = 36^\circ 4' 0''$$

$$\alpha_{P2} = 38^\circ 59' 36''$$

$$Az_{P2_P5} = 74^\circ 0' 48,44''$$

$$d_{P2_P1} = 1354,897 \text{ m}$$

$$\alpha_{P5} = 104^\circ 56' 24''$$

$$d_{P2_P5} = 825,571 \text{ m}$$

Ponto	X (m)	Y(m)
P1	7453,743	12743,125
P2	6676,216	11633,531

6. Cálculo das coordenadas de P5

$$\Delta X_{P2_P5} = d_{P2_P5} \cdot \sin Az_{P2_P5} = 793,643$$

$$\Delta Y_{AP} = d_{P2_P5} \cdot \cos Az_{P2_P5} = 227,372$$

$$X_{P5} = X_{P2} + \Delta X_{P2_P5} = 7469,859 \text{ m}$$

$$Y_{P5} = Y_{P2} + \Delta Y_{P2_P5} = 11860,903 \text{ m}$$